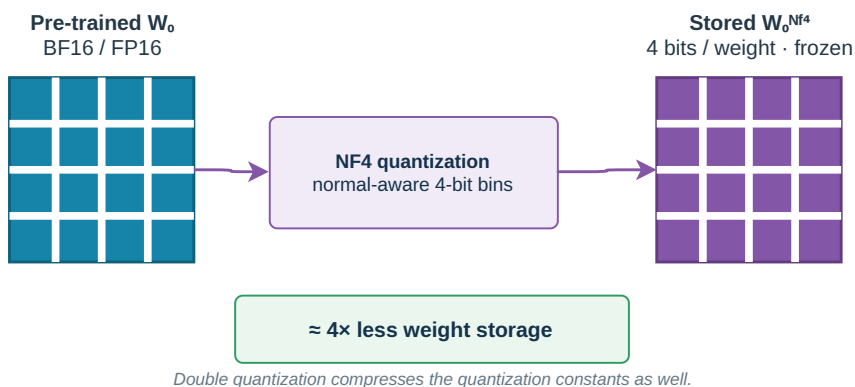


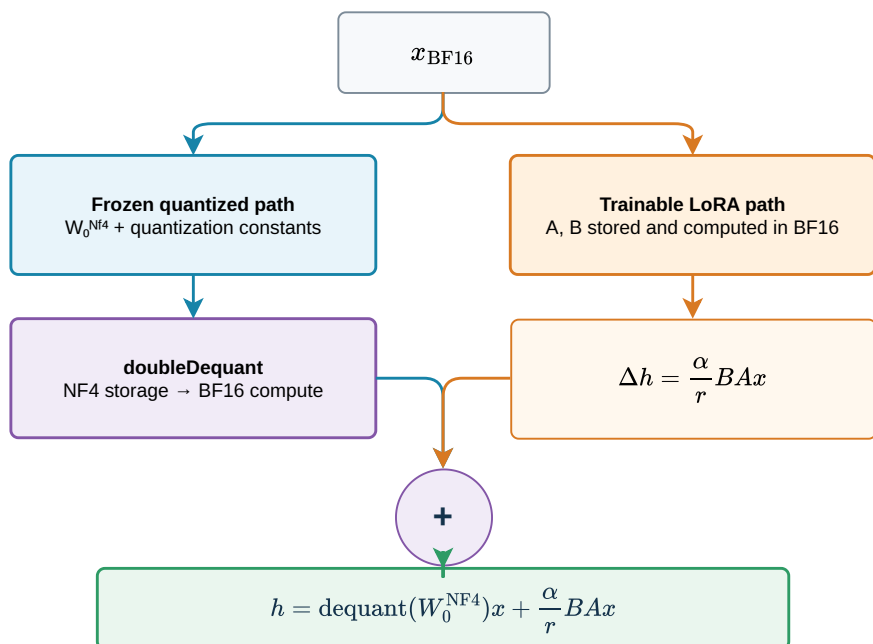
QLoRA: quantized backbone, trainable adapters

Memory-efficient fine-tuning without updating the 4-bit base weights

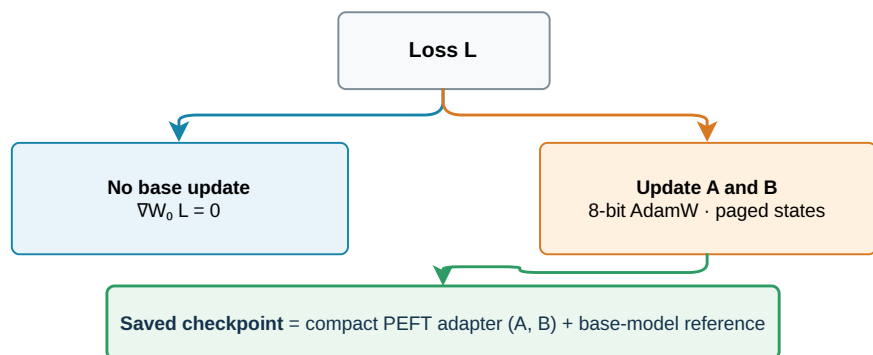
01 QUANTIZE THE FROZEN BACKBONE



02 QLORA FORWARD PASS



03 BACKPROPAGATION AND SAVED ARTIFACT



QLoRA = 4-bit frozen backbone + BF16 LoRA adapters + gradients only through A and B